Lab 2 – Light or Paint? Intrinsic Images

CV8508 – Perceptual Cognition for Computer Vision

An image is the product of two things: the *reflectance* R of the surfaces (paint) and the *shading* S of the light, $I = R \cdot S$. Lecture 4 showed that people recover R from I using cues such as penumbra, ratios across edges and co-planarity. You will build two decompositions, one from a physical prior and one learned from human judgments in Intrinsic Images in the Wild [1], test both on Adelson's checker-shadow image and on the photos where they disagree most, and then ask a recent multimodal model to do the same. About 60 minutes; a coding agent is allowed, but check what it does.

Setup

```
mkdir lab2 && cd lab2
wget https://utkarshmall13.github.io/cv8508-labs/lab2/setup.sh
sh setup.sh                       # downloads data and lab2_tools.py, builds .venv
source .venv/bin/activate
```

Read `lab2_tools.py` first (loaders, the WHDR metric, probe measurements; no decomposition code). `data/` then contains (each folder has a `README.txt` with source and licence):

`iiw` a downsized IIW subset, `train/` and `test/`. For each photo, humans compared pairs of points: "same reflectance", "left is darker" or "right is darker", with a confidence weight. `load_iiw()` returns the images and judgments. *Never train on `test`.*

`adelson` Adelson's checker-shadow image [2], with *probe* boxes on the squares A and B and the true albedo ratio of the two (`adelson.json`). The pixel values of A and B are almost the same (111 and 122 out of 255).

Tasks

Save every step as a script. Work in log space and in linear light (convert from sRGB first).

1. **Method A: physics and a simple prior, no training data (15 min).** Decompose each image into R and S using a prior you can state in one sentence for each (for example shading is smooth and reflectance is piecewise constant, the Retinex idea [3]). Report the mean WHDR on `test`. Check that your solver has converged; smooth fields converge slowly.

2. **Method B: learned from human judgments (20 min).** Train a small CNN on `train` that maps an image to log reflectance, supervised only by the human judgments (plus any prior you want). Keep it under about 500 000 parameters, train on the 160 px images for no more than 6 epochs, and expect roughly 10 minutes on a laptop CPU. Report WHDR on `test`, next to Method A and next to the trivial answer "everything has the same reflectance". *Hint:* about two thirds of the judgments say "equal", so a net that outputs log reflectance directly tends to collapse to a constant. Try predicting log shading and setting $\log R = \log I - s$ (start from $s = 0$).

3. **The Adelson image (10 min).** Run both methods on the checker-shadow image. Make one figure (image, A's reflectance, B's reflectance) and report with `probe_ratio` the three ratios listed in `adelson.json` (B/A, and the two pairs that have the same paint): in the pixels, in

the truth, from Method A and from Method B. Compute B/A once in sRGB values and once in linear light. Then, on the `test` split, find the three photos where the WHDR of A and of B differ most and show them with both reflectances.

4. **A recent multimodal model (10 min).** Give the Adelson image and your three photos to a recent multimodal model that can return images (for example the Gemini app). Ask for an albedo image and a shading image, and ask in words whether squares A and B are the same shade of gray. Save what it returns, convert it to linear light, and measure the same probe ratio on the Adelson result. Record your prompts.

Analysis

Answer in a few sentences each, citing your own numbers, and mark each claim as evidence, hypothesis or speculation.

1. How do A and B compare on WHDR, and how do both compare with the trivial answer? With 300 test images, how large a gap would be meaningful?
2. On the Adelson image, what B/A ratio did each method recover (truth and pixels are in `adelson.json`)? What did each method report for the two pairs that have the same paint, and what does that say about shadows? Which cue from the lecture does each method use, and which is missing?
3. Your ratio in sRGB values and in linear light differ. Which one is the albedo ratio, and why?
4. Look at the three photos where A and B disagree most. What is in them (shadows, textures, colored light)? Explain each disagreement from what A assumes or what B learned from human judgments.
5. Did the multimodal model's words and its picture agree? What B/A ratio did its picture give, and how did it treat your three photos? Could you tell it was wrong without the ground truth?
6. Propose one experiment on the Adelson image (for example changing the shadow edge) that would show whether a model uses shadow cues or just brightness context.

References

[1] Sean Bell, Kavita Bala, and Noah Snavely. Intrinsic images in the wild. *ACM Transactions on Graphics (SIGGRAPH)*, 33(4), 2014.
[2] Edward H. Adelson. Checker shadow illusion. https://persci.mit.edu/gallery/checkershadow.
[3] Edwin H. Land and John J. McCann. Lightness and retinex theory. *Journal of the Optical Society of America*, 61(1):1–11, 1971.